

```

#include <stdio.h>

int main() {
    int n, r, i, j, k;

    printf("Enter the Number of processes: ");
    scanf("%d", &n);
    printf("Enter the Number of resources: ");
    scanf("%d", &r);

    int alloc[n][r], max[n][r], avail[r];
    int f[n], ans[n], ind = 0;
    int need[n][r];

    printf("Enter the Allocation Matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < r; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }

    printf("Enter the MAX Matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < r; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    printf("Enter the Available Resources: ");
    for (i = 0; i < r; i++) {
        scanf("%d", &avail[i]);
    }

    for (k = 0; k < n; k++) {
        f[k] = 0;
    }

    for (i = 0; i < n; i++) {
        for (j = 0; j < r; j++) {
            need[i][j] = max[i][j] - alloc[i][j];
        }
    }

    printf("\nNeed Matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < r; j++) {
            printf("%d ", need[i][j]);
        }
        printf("\n");
    }

    for (k = 0; k < n; k++) {

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for (i = 0; i < n; i++) {
    if (f[i] == 0) {
        int flag = 0;
        for (j = 0; j < r; j++) {
            if (need[i][j] > avail[j]) {
                flag = 1;
                break;
            }
        }

        if (flag == 0) {
            ans[ind++] = i;
            for (int y = 0; y < r; y++) {
                avail[y] += alloc[i][y];
            }
            f[i] = 1;
        }
    }
}

if (ind < n) {
    printf("\nNo safe sequence found. The system is in an unsafe state.\n");
} else {
    printf("\nThe SAFE Sequence is as follows:\n");
    for (i = 0; i < n - 1; i++) {
        printf(" P%d ->", ans[i]);
    }
    printf(" P%d\n", ans[n - 1]);
}

return 0;
}

```

```
csec2@sicet-Veriton-M200-H410:~/Goku1$ ./a.out
```

```
Enter the Number of processes: 3
```

```
Enter the Number of resources: 3
```

```
Enter the Allocation Matrix:
```

```
1
```

```
0
```

```
2
```

```
4
```

```
2
```

```
1
```

```
0
```

```
4
```

```
1
```

```
Enter the MAX Matrix:
```

```
3
```

```
4
```

```
1
```

```
2
```

```
3
```

```
0
```

```
1
```

```
2
```

```
3
```

```
Enter the Available Resources: 2
```

```
1
```

```
3
```

```
The SAFE Sequence is as follows:
```

```
P1 -> P2 -> P0
```

```
csec2@sicet-Veriton-M200-H410:~/Goku1$
```